

CV - Aniruddha Marathe, PhD

aniruddha.pravin.marathe@gmail.com | (970)-234-4161 | <https://www.linkedin.com/in/aniruddha-marathe-phd>

PROFESSIONAL SUMMARY

- Multidisciplinary researcher with 10 years of experience in community ecology and macro-ecology, and conservation as evidenced by 17 publications in journals such as *PLOS-One*, *Oryx*, and *Evolution*.
- Demonstrated ability to collaborate, design research, and manage multidisciplinary projects, exemplified by leadership in three research endeavors focused on community assembly and continental scale biodiversity patterns and impacts of climate change.
- Proficient in developing data driven analytical solutions for biodiversity research, as evidenced by the R package 'rangemodLR' (<https://cran.r-project.org/package=rangemodLR>)

EDUCATION

- **Ph.D. | Manipal University, India** | October 2018
Dissertation: Species richness and distribution of ants along an elevational gradient in the Eastern Himalaya
- **M.Sc. | University of Pune, India** | July 2009
Dissertation: Comparison of species composition of ants across different habitats near the Pune city

Technical Skills

- **Analytical:** Study design | Ecological modeling | multivariate statistics | species distribution models | geospatial analysis | Excel | data wrangling |
- **Programming:** R | R Markdown | R Shiny | Python |
- **GIS:** ArcGIS | QGIS | Introductory knowledge Google Earth Engine
- **Field and laboratory skills:** Systematic sampling of invertebrates | Management of museum collections | Ant identification & taxonomy | Molecular laboratory protocols | Microscopic photography and imaging

Grants and Funding

- CSIR- Research Associate fellowship-2021
- Critical Ecosystem Partnership Fund - small grant, 2011

EXPERIENCE

Research Associate | Quantitative analysis | GIS | Project Management

Center for Cellular and Molecular Biology, Hyderabad, India | 2023 - 2024

- Managed research projects related to understanding the biogeography, species distributions, and genetic diversity of centipedes in peninsular India
- Planned and executed field expeditions for systematic sampling of centipedes, and other soil arthropods leveraging knowledge from GIS, and natural history
- Designed and coordinated protocols for downstream processing of metadata for specimens, and other laboratory protocols
- Generated scientific insights from the field data using species distribution models, ecological niche analysis, and regression models
- Outputs of this work is under review in the journal *Toxcon*

Postdoctoral Research Scientist | Species distribution models | Biogeography | Climate change

Indian Institute of Science, Bengaluru, India | 2019 - 2023

- Quantified the climate change extinction risk for key biodiversity components endemic to the western ghats biodiversity hotspot using species distribution models and regression analysis
- Identified key habitat areas at global scale for seven species of critically endangered marine turtles
- Developed automated workflows in R for predicting geographic distributions of 132 species of frogs, and 28 species of lizards
- Outputs of the work are published in the journal *Evolution* and in review in *PLOS-One*

Postdoctoral Research Associate | Biodiversity Inventory | Data curation

Ashoka Trust for Research in Ecology and the Environment, Bengaluru, India | 2018-2019

- Facilitated cross institutional collaboration to build biodiversity database for regions of global conservation importance within the Himalaya
- Deployed shiny apps dashboard for data visualization and coordination across partners
- Trained a cohort of 9 postdocs, and 18 research fellows in good practices of data curation and sharing

Peer reviewed Publications

- Link to google scholar profile: <https://scholar.google.com/citations?user=WAUNZ7oAAAAJ&hl=en>

Recent and upcoming publications

Marathe, A., Sahanashree, R., Priyadarsanan, D. R. (In review). Two new species of *Tetramorium* Mayr, 1855 (Hymenoptera: Formicidae: Myrmicinae) from Northeast India. *Asian Myrmecology*.

Marathe A., S. P. Vijaykumar, Torsekar V, K. P. Dinesh, Pal S., Achyuthan S., Shanker K. (In review). Range and elevation predict responses to climate change in frogs and lizards in the Western Ghats biodiversity hotspot of peninsular India. *PLOS ONE*.

Aditi, Roy, P., Parikh, R., **Marathe, A.**, Majhi, K., Jenner, R., Joshi, J., 2026. Divergent venoms among two closely related co-distributed centipede species, *Scolopendra morsitans* and *S. hardwickei* in tropical Asia. *bioRxiv* 2026–04.

Shanker K., Torsekar V, S. P. Vijaykumar, K. P. Dinesh, **Marathe A.**, Cyriac V. (2026). Body size evolution and the co-occurrence of stream-dwelling frogs in the Western Ghats of India. *Evolution* *qpag060*

Ghosh A., **Marathe A.**, Cyriac V., Umesh P., Kulkarni V., Datta-Roy A. (2024). A Lankan invasion? An intriguing case of the presence of a Sri Lankan endemic skink on the southwestern Indian Coast. *Systematics and Biodiversity* 22(1): 2383215. <https://doi.org/10.1080/14772000.2024.2383215>

Sahanashree, R., Aswaj, P., **Marathe, A.**, Priyadarsanan, D. R. (2023) Two new reports of *Stigmatomma* Roger, 1859 (Hymenoptera, Formicidae, Amblyoponinae) species from the Indian subcontinent. *Sociobiology* 70(4), 39315

Conference presentations

Marathe A., Sudhanshu K., Dhanush SS, Joashi J. (2024). Predictors of genetic diversity in an ancient predatory soil arthropod group (Scolopendromorpha) in peninsular India. Indian Wildlife Ecology Conference, Bangalore - June 2024

Marathe A., S.P. Vijayakumar, Varun Torsekar , K.P. Dinesh and Kartik Shanker (2023). Latitudinal and elevational range shifts under future climate change in the Western Ghats frogs. Association for Tropical Biology and Conservation conference, Coimbatore, 2 – 6 July 2023

Marathe A and Priyadarsanan, D. R., (2010). Exploration of ant diversity on elevation gradient in Eaglenest wildlife sanctuary (NE Himalayas, Arunachal Pradesh), Indian Biodiversity Congress (IBC 2010), organized by CISSA in Thiruvananthapuram, 27-31 December 2010

Marathe A and Priyadarsanan D. R. , 2015. "Potential effects of, warming on ant species distributions across elevation gradient in the Eastern Himalaya", The Future of Biodiversity in Tropical Asia: addressing local and global challenges, Organized by the Association of Tropical Biology and Conservation (ATBC) at Phnom Penh, Cambodia 30-March to 2-April 2015

Other academic achievements and contributions

- National Eligibility Certificate (CSIR-NET), 2007. (F.No.10-2(5)/2007 (ii)-E.U.II)
- Participated in the 'Ant Course' in 2010 conducted by the California Academy of Sciences
- Conducted workshops on exploratory data analysis in the Student Conference on Conservation Science - Bangalore (SCCS- Bangalore, 2017) and ecological data management at SCCS- Bangalore, 2018
- Code in place python programming course, Stanford University, CA